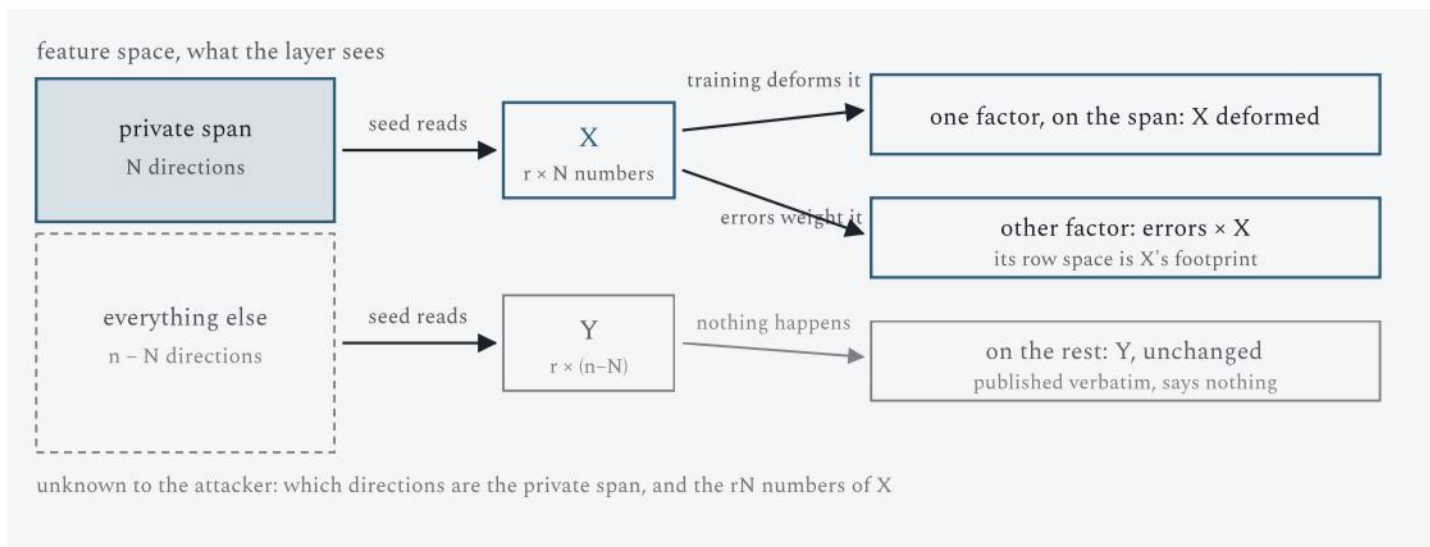




## 'motorcycle' added as a new class to a CIFAR-10 MLP

135 of 200 random starts recover a private image; 8 of 8 images found

bottom two rows are the two ends of the attacker's OWN ranking by residual — no ground truth is used to sort them

private image



chart projection  
(the fidelity ceiling)



recovered  
residual  $3e-29$

recovered  
residual  $2e-28$

recovered  
residual  $6e-29$

recovered  
residual  $4e-29$

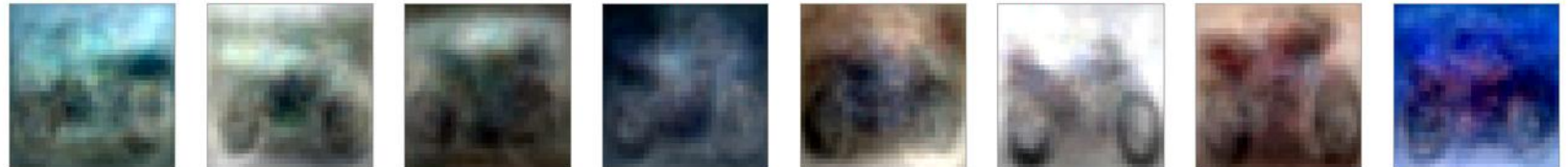
recovered  
residual  $2e-29$

recovered  
residual  $4e-29$

recovered  
residual  $2e-29$

recovered  
residual  $2e-29$

BEST-scoring starts  
(what the attacker keeps)



'fashion\_bag' as an 11th class on a CIFAR-10 CNN (92.7% test / 99.83% train) -- head LoRA  $r=128$ ,  $T=400$  SGD steps ( $lr=0.01$ ), public PCA chart  $k=64$ , onchart privates  
 9/150 random starts landed, 4/8 images found, residual at truths  $1e-13$  vs public median  $3.9e-01$

private (raw)

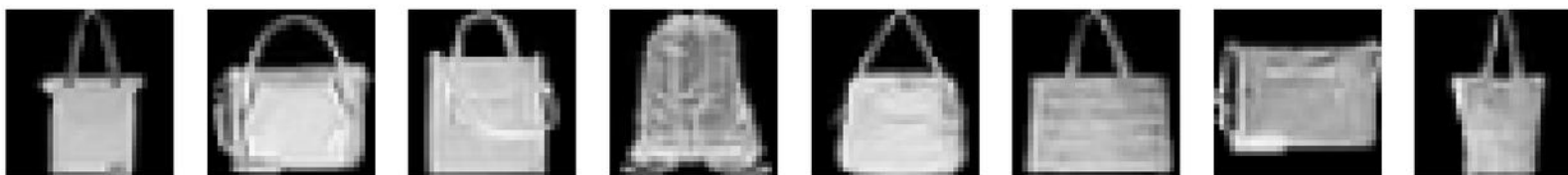
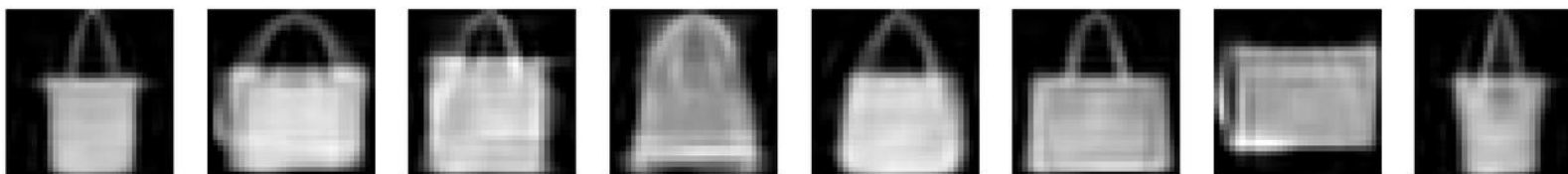


chart projection (public PCA)



err 0.35  
ssim 0.40

landed  
ssim 0.65

landed  
ssim 0.60

err 0.39  
ssim 0.36

err 0.20  
ssim 0.54

landed  
ssim 0.70

landed  
ssim 0.74

-

closest random start (certificate only)

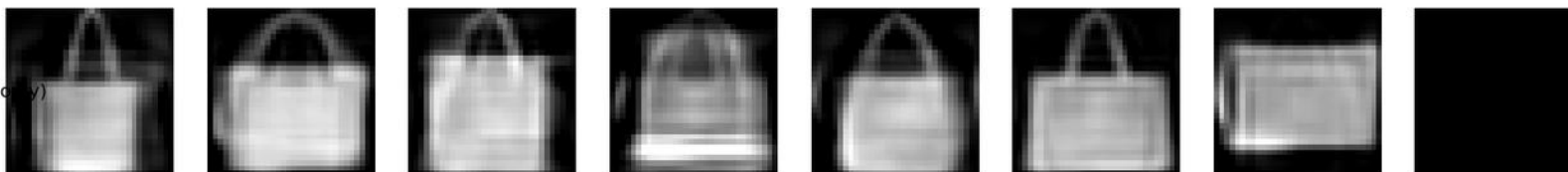
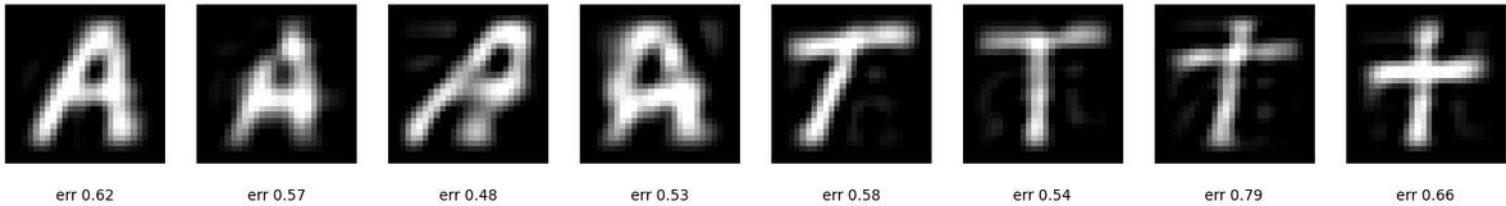
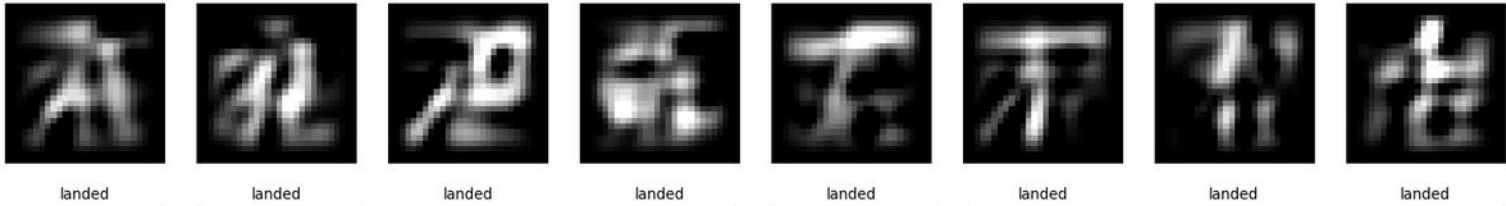


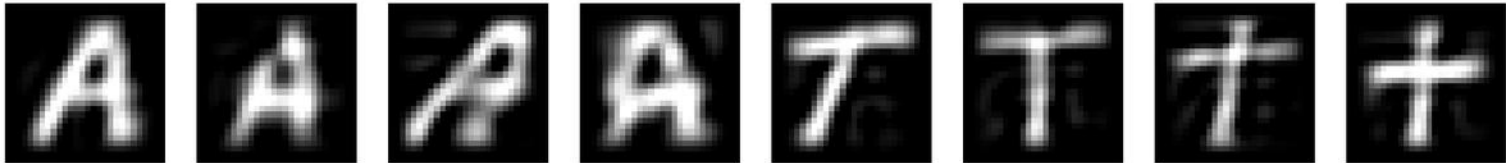
chart projection (public PCA k=32)



NTK linearised (free coefficients)



Certificate (native Levenberg-Marquardt)



## the private photograph



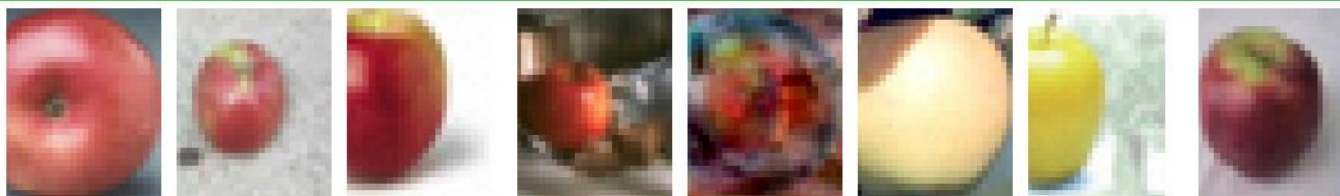
## chart contains the photographs

191/400 land · 7/8 found · SSIM 0.90 (ceiling 1.00)



## chart 2% off

169/400 land · 4/8 found · SSIM 0.89 (ceiling 0.99)



## chart 5% off

0/400 land · SSIM 0.85 (ceiling 0.94)



## chart 10% off

0/400 land · SSIM 0.76 (ceiling 0.84)



## chart 20% off

0/400 land · SSIM 0.62 (ceiling 0.67)



## PUBLIC PCA chart (the attacker's actual chart)

0/400 land · SSIM 0.35 (ceiling 0.58) · control 0.51



Same capacity boundary, three charts — BELOW the line ( $k = 17 < 18$ )  
 all three collapse at  $k = m+r-N = 18$ ; what comes back depends entirely on the chart.  
 The third chart is ORACLE-BUILT from the private digits: a control on what the count can see, NOT an attack.

